

Honors Complex Analysis

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1 Introduction to complex numbers

1.1 Basic definitions

Definition 1. The set of complex numbers, $\mathbb{C}$, is defined by

$$\mathbb{C} := \{x + yi | x, y \in \mathbb{R}\},$$

where $i^2 = -1$.

Like real numbers, $\mathbb{C}$ is equipped with basic operations. Let $z = a + bi, w = c + di$, then

- $z + w = (a + bi) + (c + di) = (a + c) + (b + d)i$,
- $zw = (a + bi)(c + di) = (ac - bd) + (bc + ad)i$.
- We have $|z \cdot w| = |z| \cdot |w|$. Verify it yourself.
- Suppose $(a, b) \neq (0, 0)$, then the *inverse* of z is defined by

$$z^{-1} = \frac{1}{z} = \frac{1}{a + bi} = \frac{(a - bi)}{(a + bi)(a - bi)} = \frac{1}{a^2 + b^2}(a - bi).$$

- The *complex conjugate* of $z = a + bi$ is denoted by $\bar{z} = a - bi$, the *norm* of $z = a + bi$ is defined by $|z| = \sqrt{a^2 + b^2}$, hence

$$z \cdot \bar{z} = |z|^2.$$

We may also represent a complex number using polar coordinates:

$$z = a + bi = r(\cos \theta + i \sin \theta),$$

where $r = \sqrt{a^2 + b^2}$, $\theta = \arctan(b/a)$. We define $\arg(z) = \theta$.

Polar representation is useful when multiplying complex numbers. Suppose $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$\begin{aligned} z_1 \cdot z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 \left((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \right) \\ &= r_1 r_2 \left(\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) \right). \end{aligned}$$

That is, we have

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2).$$

Theorem 1. $\cos \theta + i \sin \theta = e^{i\theta}$.

When $\theta = \pi$, then we have the famous Euler's formula:

$$e^{i\pi} + 1 = 0.$$

The number $e \approx 2.7182818 \dots$ is an irrational number defined through the limit

$$e = \lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right)^N.$$

One can also show that

$$\begin{aligned}
e^x &= \left(\lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right) \right)^x \\
&= \lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right)^{Nx} \text{ by Monotone convergence theorem,} \\
&= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{N} \right)^N \text{ by Bernoulli inequality.}
\end{aligned}$$

Hence we have

$$e^{i\theta} = \lim_{N \rightarrow \infty} \left(1 + \frac{i\theta}{N} \right)^N.$$

As $N \rightarrow \infty$, small angle approximation yields

$$\begin{aligned}
1 + i \frac{\theta}{N} &\approx \cos \left(\frac{\theta}{N} \right) + i \sin \left(\frac{\theta}{N} \right), \\
\left(1 + i \frac{\theta}{N} \right)^N &\approx \left(\cos \left(\frac{\theta}{N} \right) + i \sin \left(\frac{\theta}{N} \right) \right)^N.
\end{aligned}$$

1.2 Fundamental theorem of algebra

Theorem 2. *If $f(z) = a_n z^n + \dots + a_1 z + a_0$ is a polynomial with complex coefficients, then $\exists z \in \mathbb{C}$ such that $f(z) = 0$.*

First proof. If $|z| \gg 0$, $|z| = R > 1$, then $|a_n z^n| = |a_n| \cdot R^n$,

$$\begin{aligned}
|a_{n-1} z^{n-1} + \dots + a_1 z + a_0| &\leq |a_{n-1}| \cdot R^{n-1} + \dots + |a_1| \cdot R + |a_0| \\
&\leq R^{n-1} \cdot (|a_0| + |a_1| + \dots + |a_{n-1}|).
\end{aligned}$$

Let z_0 be a point inside the open disc $|z| = R$ such that $|f(z_0)|$ is minimized. By transforming z_0 to 0, we may assume $|f(0)|$ is the minimum, and $|f(0)| = |a_0|$. Now suppose after transformation, the polynomial is

$$f(z) = a_0 + a_k z^k + \dots + a_n z^n = a_0 + a_k z^k (1 + a_{k+1} z + \dots + a_n z^{n-k})$$

where a_k is the first non-zero coefficients. Choose $z = \left(-\frac{a_0}{a_k} \right)^{1/k} \epsilon$ for some small $\epsilon > 0$, then

$$\begin{aligned}
f(z) &= a_0 - a_0 \epsilon^k \left(1 + \frac{a_{k+1}}{a_k} z + \dots + \frac{a_n}{a_k} z^{n-k} \right) \\
&\approx a_0 (1 - \epsilon k),
\end{aligned}$$

which implies $|f(z)| < |f(0)|$, a contradiction to the minimality of $|f(0)|$, unless $a_0 = 0$. Hence in original polynomial $f(z_0) = 0$, which is a root. $\square$

Topological proof. Wlog, suppose $f(z) = z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$. When $|z|$ is large, $f(z)$ is dominated by z^n , and

$$|f(z) - z^n| = |a_{n-1} z^{n-1} + \dots + a_1 z + a_0| < C \cdot |z|^{n-1},$$

where C is some constant not depending on z . Let $|z| = R$ be the circle with radius R , then z^n is the circle with radius R^n and winding n times. The remaining term $a_{n-1}z^{n-1} + \dots + a_0$ will perturbate around $z^n = R^n$. When $R = 0$, the path is concentrated around a_0 . For some R , the path must cross 0, indicating a root. $\square$

Using Liouville's theorem. We claim that any bounded differentiable function $f : \mathbb{C} \rightarrow \mathbb{C}$ is constant (Liouville's theorem). Suppose $p(z) \in \mathbb{C}[z]$ is a polynomial with no roots, then let $f(z) = \frac{1}{p(z)}$, then

$$\frac{d}{dz} \frac{1}{p(z)} = -\frac{p'(z)}{p^2(z)},$$

and by Liouville's theorem, $f(z)$ is a constant, so is $p(z)$. $\square$

1.3 Topological structures of $\mathbb{C}$

Definition 2. A sequence of complex numbers $\{z_n\} \subset \mathbb{C}$ is said to converge to $z \in \mathbb{C}$, if $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$, such that $d(z_n, z) < \epsilon$ for all $n \geq N$.

The distance is the norm of the difference: $d(z_n, z) = |z_n - z|$, since $\mathbb{C}$ is a metric space.

Definition 3. A sequence $\{z_n\} \subset \mathbb{C}$ is Cauchy, if $\forall \epsilon > 0$, $\exists N \subset \mathbb{N}$, such that $|z_k - z_l| < \epsilon$, $\forall k, l \geq N$.

Note that $\mathbb{C}$ is complete, meaning every Cauchy sequence in $\mathbb{C}$ converges.

Definition 4. We define the following sets in $\mathbb{C}$:

- The open disc with radius r centered at z_0 : $D_r(z_0) := \{z \in \mathbb{C}, |z - z_0| < r\}$.
- The closed disc with radius r centered at z_0 : $\bar{D}_r(z_0) := \{z \in \mathbb{C}, |z - z_0| \leq r\}$.
- The boundary of a closed disc: $C_r(z_0) := \{z \in \mathbb{C}, |z - z_0| = r\}$.

For special notation, the unit disc is denoted by $\mathbb{D} := D_1(0)$.

Definition 5. Given $\Omega \subset \mathbb{C}$, a point $z_0 \in \Omega$ is an interior point if $\exists r > 0$, such that $D_r(z_0) \subset \Omega$. A set $\Omega \subset \mathbb{C}$ is open, if $\forall z \in \Omega$, z is an interior point. A set $\Omega \subset \mathbb{C}$ is closed if its compliment, $\bar{\Omega}$, is open.

The collection of all open sets in $\mathbb{C}$, $\{\mathcal{U}_i\}$ forms a *topology* on $\mathbb{C}$, i.e.,

- $\emptyset, \mathbb{C}$ are open;
- $\bigcap_{i=1}^n \mathcal{U}_i$ is open. We say it is closed under finite intersection;
- $\bigcup_{\mathcal{I}} \mathcal{U}_i$ is open. We say it is closed under arbitrary union.

Definition 6. A point $z \in \mathbb{C}$ is a limit point for $\Omega \subset \mathbb{C}$, if there exists a sequence $\{z_n\} \subset \Omega$, and

$$\lim_{n \rightarrow \infty} z_n = z.$$

Proposition 1. If Ω is closed, then every limit point of Ω lies in Ω .

Proof. Let $z \in \mathbb{C}$, $z = \lim_{n \rightarrow \infty} z_n$ where $z_n \in \Omega$. Suppose $z \notin \Omega$, then $z \in \bar{\Omega}$ open, so $\exists r > 0$, such that $D_r(z) \subset \bar{\Omega}$. But choose $\epsilon < r$, we get a contradiction. $\square$

Definition 7. Given $\Omega \subset \mathbb{C}$, define $\text{Int}(\Omega)$ as the set of interior points of Ω , $\bar{\Omega}$ is defined as the closure of Ω , which is the set of limit points of Ω . We have

$$\text{Int}(\Omega) \subset \Omega \subset \bar{\Omega}.$$

Finally, the boundary of Ω , denote as $\partial\Omega$, is defined by $\partial\Omega := \bar{\Omega} - \text{Int}(\Omega)$.

Definition 8. A set $\Omega \subset \mathbb{C}$ is bounded, if $\exists R > 0$, such that $\Omega \subset D_R(0)$.

Definition 9. A set $\Omega \subset \mathbb{C}$ is compact, if it is both closed and bounded.

Here definition 9 is not rigorous. The formal definition of compactness says that a set is compact if every open cover has a finite subcover. We use the Heine-Borel theorem to conclude $\Omega \subset \mathbb{R}^n$ is compact iff it is closed and bounded, and use $\mathbb{R}^2 \simeq \mathbb{C}$.

Definition 10. A set $\Omega \subset \mathbb{C}$ is sequentially compact, if every sequence $\{z_n\} \subset \mathbb{C}$ has a convergent subsequence.

Proposition 2. If a set $\Omega \subset \mathbb{C}$ is compact, then Ω is sequentially compact.

Proof. Since Ω is compact, every open cover has a finite subcover. For each $k \geq 1, k \in \mathbb{N}$, consider the families of open covers defined by $\{B_{2^{-k}}(w), w \in \Omega\}$. Then, for each k , we have a finite collection of complex numbers $w_{k_1}, \dots, w_{k_{m_k}}$ such that

$$\Omega \subset \bigcup_{i=1}^{m_k} B_{2^{-k}}(w_{k_i}).$$

Let $\{z_n\} \subset \Omega$ be any sequence. When $k = 1$, Ω is covered by a finitely many balls of radius $1/2$. At least one of those balls contains infinitely many terms in $\{z_n\}$, so let $z_{n_1}^{(1)} \subset z_n$ be a subsequence such that $z_{n_1}^{(1)} \in B_{1/2}(\cdot)$, i.e., all terms in $\{z_{n_1}^{(1)}\}$ lie in the same ball with radius $1/2$. Using the induction idea, suppose we have $\{z_{n_k}^{(k)}\}$ chosen, then we can construct a subsequence $z_{n_{k+1}}^{(k+1)} \subset z_{n_k}^{(k)}$, such that all terms of $z_{n_{k+1}}^{(k+1)}$ lie in the same ball with radius $2^{-(k+1)}$. Now define $y_k = z_{n_k}^{(k)}$, by our construction for all k , when $m \geq k$ $y_m \in B_{2^{-k}}(\cdot)$. Hence

$$|y_m - y_n| < 2 \cdot 2^{-k} \rightarrow 0, \quad \forall m, n \geq k.$$

Hence we conclude $\{y_k\}$ is Cauchy. Since $\mathbb{C}$ is complete, so denote

$$\lim_{k \rightarrow \infty} y_k = y.$$

Finally since Ω is closed and bounded, $y \in \Omega$. Hence every sequence has a convergent subsequence. $\square$

Definition 11. A set $\Omega \subset \mathbb{C}$ is connected, if Ω cannot be written as the union of open sets. $\Omega \subset \mathbb{C}$ is called a region if it is open and connected.

2 Holomorphic functions

2.1 Cauchy-Riemann equations

Definition 12. A function $f(z) : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic if it has a well defined derivative, i.e.,

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \text{ exists and is well defined.}$$

Let $f(z) = f(x+iy) = u(x, y) + iv(x, y)$, and suppose f is holomorphic. We compute the derivative $f'(z)$ in two ways: First, let $h \in \mathbb{R}$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} &= \lim_{h \rightarrow 0} \frac{u(x+h, y) + iv(x+h, y) - u(x, y) - iv(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x+h, y) - u(x, y)}{h} + i \lim_{h \rightarrow 0} \frac{v(x+h, y) - v(x, y)}{h} \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \end{aligned} \tag{2.1}$$

Second, let $h \in \mathbb{R}$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+ih) - f(z)}{ih} &= \lim_{h \rightarrow 0} \frac{u(x, y+ih) + iv(x, y+ih) - u(x, y) - iv(x, y)}{ih} \\ &= \lim_{h \rightarrow 0} \frac{u(x, y+ih) - u(x, y)}{ih} + i \lim_{h \rightarrow 0} \frac{v(x, y+ih) - v(x, y)}{ih} \\ &= \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \end{aligned} \tag{2.2}$$

Since f is holomorphic, by (2.1) and (2.2), we have the *Cauchy-Riemann equation*:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}}. \tag{2.3}$$

If $f = (u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, the Jacobian of f is defined as

$$\mathcal{J}_f = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \stackrel{(2.3)}{=} \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad a, b \in \mathbb{R}.$$

Note that if we let $z = a + bi$ and define a map T by $w \mapsto (a + bi)w$ where $w \in \mathbb{C}$. Take the basis of $\mathbb{C}$ as $(1, 0), (0, i)$, then the transformation matrix is given by

$$T(w) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} w.$$

Think of $z = \cos \theta + i \sin \theta$, we hence have the transformation matrix:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which is the counter-clockwise rotation by θ .

Theorem 3. *If a function $f(z) : \mathbb{C} \rightarrow \mathbb{C}$ satisfies the Cauchy-Riemann equation, then $f(z)$ is holomorphic.*

Proof. Let $h = h_1 + ih_2$, $f(z) = u(x, y) + iv(x, y)$. By linear approximation, we have

$$u(x + h_1, y + h_2) = u(x, y) + \frac{\partial u}{\partial x}h_1 + \frac{\partial u}{\partial y}h_2 + ||h|| \cdot \psi_1(h), \quad (2.4)$$

where $\psi_1(h) \rightarrow 0$ as $h \rightarrow 0$. Likewise,

$$v(x + h_1, y + h_2) = v(x, y) + \frac{\partial v}{\partial x}h_1 + \frac{\partial v}{\partial y}h_2 + ||h|| \cdot \psi_2(h), \quad (2.5)$$

where $\psi_2(h) \rightarrow 0$ as $h \rightarrow 0$. Then combine (2.4) and (2.5) and use Cauchy-Riemann equation, we have

$$f(z + h) = f(z) + \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) (h_1 + ih_2) + ||h|| \cdot \psi(h),$$

where $\psi(h) \rightarrow 0$ as $h \rightarrow 0$, hence $f'(z)$ exists. □

2.2 Examples of holomorphic functions

We investigate some holomorphic functions:

① $f(z) = z^n$. We have that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} &= \lim_{h \rightarrow 0} \frac{(z+h)^n - z^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} z^{n-k} h^k}{h} \\ &= \lim_{h \rightarrow 0} n z^{n-1} + \binom{n}{2} h z^{n-2} + \dots + h^{n-1} \\ &= n z^{n-1}. \end{aligned}$$

We have $f'(z) = n z^{n-1}$.

② If $f(z), g(z)$ are holomorphic, then so is

$$\lambda_1 f(z) + \lambda_2 g(z), \quad \lambda_1, \lambda_2 \in \mathbb{C},$$

and

$$(\lambda_1 f(z) + \lambda_2 g(z))' = \lambda_1 f'(z) + \lambda_2 g'(z).$$

Based on this property, $f(z) \in \mathbb{C}[z]$ is holomorphic.

③ Power series: A power series is an infinite sum

$$\sum_{n=0}^{\infty} a_n z^n, \quad a_n \in \mathbb{C},$$

we say the series converges at $z_0 \in \mathbb{C}$ if

$$\lim_{N \rightarrow \infty} \sum_{n=0}^N a_n z_0^n \text{ exists.}$$

We will see later that, if the power series converges absolutely in $\Omega \subset \mathbb{C}$, then it defines a holomorphic function.

④ A function that is not holomorphic: $f(z) = \bar{z}$. Derivative differ if you approach from real and imaginary axis.

2.3 Power series

As we have shown, polynomials are holomorphic. Indeed, holomorphic functions form a ring.

Definition 13. A power series is a formal expression of the form

$$\sum_{n=0}^{\infty} a_n z^n, a_n \in \mathbb{C}.$$

If $z \in \mathbb{C}$, we say that the series $\sum a_n z^n$ converges if

$$\lim_{N \rightarrow \infty} \sum_{n=0}^N a_n z^n \text{ converges.}$$

We say $\sum a_n z^n$ converges absolutely, if

$$\sum_{n=1}^{\infty} |a_n| \cdot |z^n| \text{ converges.}$$

Theorem 4. If $\sum a_n z^n$ convergence absolutely, then $\sum a_n z^n$ converges.

Proof. Note that for sufficiently large N ,

$$\left| \sum_{n=N}^{\infty} a_n z^n \right| \leq \sum_{n=N}^{\infty} |a_n| \cdot |z^n| < \epsilon.$$

□

Theorem 5. Given a power series $\sum a_n z^n$, $\exists 0 \leq R \leq \infty$, such that the following properties hold:

(i) If $|z| < R$, then $\sum a_n z^n$ converges absolutely;

(ii) If $|z| > R$, then $\sum a_n z^n$ diverges.

Furthermore, $L = \frac{1}{R} = \limsup |a_n|^{1/n}$. R is called the radius of convergence.

Proof. Let $L = \frac{1}{R}$, suppose $|z| < R$, then $\exists \epsilon > 0$, such that $r := (L + \epsilon) \cdot |z| < LR < 1$, and $\exists N$ such that $L + \epsilon > |a_n|^{1/n}$, $\forall n \geq N$. Hence

$$|a_n|^{1/n} \cdot |z| < (L + \epsilon) \cdot |z| < 1 \implies |a_n| \cdot |z|^n < r^n.$$

Hence $\sum |a_n| \cdot |z|^n$ converges by comparing test with $\sum r^n, r < 1$. On the other hand, if $|z| > R$, then there are infinitely many n for which $|a_n|^{1/n} > \frac{1}{R} - \epsilon$, so

$$|a_n|^{1/n} \cdot |z| > 1 \implies |a_n| \cdot |z|^n > r^n,$$

and $\sum |a_n| \cdot |z|^n$ diverges by comparing test.

□

Example 1. Consider the radius of convergence for the following series:

- (i) $\sum n!z^n$, $R = 0$.
- (ii) $\sum \frac{z^n}{n!}$, $R = \infty$.
- (iii) $\sum z^n$, $R = 1$.

Theorem 6. A power series $f(z) = \sum a_n z^n$ admits a derivative on its disc of convergence, and $f'(z) = \sum n a_n z^{n-1}$ with the same radius of convergence.

Proof. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$, $g(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$, and write

$$f(z) = \underbrace{\sum_{n=0}^N a_n z^n}_{S_N(z)} + \underbrace{\sum_{n=N+1}^{\infty} a_n z^n}_{E_N(z)}.$$

Fix $z_0 \in D_R(0)$,

$$\begin{aligned} \left| \frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) \right| &= \left| \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) + S'_N(z_0) - g(z_0) + \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| \\ &\leq |A| + |B| + |C|, \end{aligned}$$

where

$$\begin{aligned} |A| &= \left| \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right|, \\ |B| &= |S'_N(z_0) - g(z_0)|, \quad |C| = \left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right|. \end{aligned}$$

Note that $\forall \epsilon > 0, \exists N_1$, such that

$$|S'_N(z_0) - g(z_0)| = \left| \sum_{n=N}^{\infty} n a_n z^n \right| < \epsilon, \forall N \geq N_1, \quad (2.6)$$

and moreover

$$\begin{aligned} \left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| &\leq \sum_{n=N+1}^{\infty} |a_n| \cdot \frac{(z_0 + h)^n - z_0^n}{h} \\ &= \sum_{n=N+1}^{\infty} |a_n| \cdot \frac{h((z_0 + h)^{n-1} + z_0(z_0 + h)^{n-2} + \dots + z_0^{n-1})}{h} \\ &\leq \sum_{n=N+1}^{\infty} |a_n| \cdot n \cdot R^{n-1}. \end{aligned}$$

Hence $\exists N_2$, such that

$$\left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| < \epsilon, \forall N \geq N_2.$$

Note that $\exists \delta$, such that $\forall |h| < \delta$, $|A| < \epsilon$ by the definition of derivative. Hence take $N \geq \max\{N_1, N_2\}$, one have

$$\left| \frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) \right| < 3\epsilon.$$

□

Corollary 1. $f(z) = \sum a_n z^n \in C^\infty(D_R(0))$.

More generally, we write $\sum a_n (z - z_0)^n$ converges for $z \in D_R(z_0)$.

Definition 14. A function $f : \Omega \rightarrow \mathbb{C}$, where Ω is a region, is analytic, if it admits a power series on $D_R(z_0)$ for all $z_0 \in \Omega$.

We shall see that, f is holomorphic $\iff f$ is analytic.

3 Complex integrals

3.1 Integration along curves

Definition 15. A parametrized curve is a function $\gamma : [0, 1] \rightarrow \mathbb{C}$, where γ is differentiable with continuous non-zero derivative.

Definition 16. 2 parametrization $\gamma_1, \gamma_2 : [0, 1] \rightarrow \mathbb{C}$ are equivalent if $\exists s : [0, 1] \rightarrow [0, 1]$ smooth, $s(0) = 0, s(1) = 1$, such that $\gamma_2 = \gamma_1 \circ s$.

Definition 17. If γ is a parametrized curve, then

$$\int_{\gamma} f(z) dz = \int_0^1 f(\gamma(t)) \cdot \gamma'(t) dt.$$